apparently reduce oil import costs by ₹3.5 lakh crores, save 846 million tons of net CO2 emissions, and oil savings of 474 MTOE. However, the present ground reality is that EV penetration in India stands at just about 0.8% and only 4% of that is four-wheelers.

According to the Electric Vehicle Outlook 2022 by BloombergNEF, there are now 20 million passenger EVs, 1.3 million commercial EVs, and over 280 million two and three-wheelers. China continues to dominate the global EV market followed by Europe. Despite being projected to grow at a CAGR of 90% between 2021 and 2030, several challenges in EV adoption have been responsible for the present sluggishness in the Indian market.

For context, here are some statistics according to the Accelerated e-Mobility Revolution for India's Transportation (e-Amrit) portal at the time of writing:

- No. of EVs registered: 759,182
- No. of EV manufacturers: 380
- No. of installed charging stations: 1,800
- No. of EV sales in FY21-22: 1.32%

These numbers clearly indicate that EV penetration is still quite low in the Indian market. However, things are not all gloomy. A KPMG market survey does point to an uptick in EV sales in FY20 compared to FY19.

Leading the EV bandwagon are three-wheeler EVs, which constituted 14.1% (90,000 units) of the total vehicles sold in this segment. Two-wheelers form the bulk of transport vehicles in India, but only 0.9% (152,000) of them were EVs in FY20.

Four-wheeler passenger vehicles have shown extremely low penetrance at just 0.12% (3,400) of the units sold in FY20. Reasons for this include large pricing disparities when compared to their internal combustion engine (ICE) equivalents, no standardization in range, and inadequate charging infrastructure. Dearth of public parking, long charging times, and difficulty in battery swapping are likely to hinder growth of the four-wheeler EV segment.

Commercial vehicles such as buses accounts for the least number of EVs sold in India. In 2020, only 600 of the 515,600 EVs sold in India were e-buses.

The commercial vehicle industry suffers from its own challenges including but not limited to import dependency for the chassis and other components not to forget the battery, lack of public charging infrastructure, and the lack of standardization in battery capacity and life estimates during commutes to name a few.

However, commercial vehicles such as intra-city buses have the highest propensity to transition to EVs. Intra-city buses have a high daily run and operate in predictable routes. Most Indian cities with a sizeable population have multiple bus depots, which can be utilised as charging points. The overall cost savings in terms of fuel can be passed on the consumer, which can result in cheaper fares for long distances or even in AC bus travel.

The two-wheeler segment has the highest number of vehicles on the road. However, the propensity for an

average commuter to change his/her ICE two-wheeler to an EV is comparatively lower than what is possible in the case of commercial vehicles. That being said, two-wheelers come with the added advantage of being able to have charging equipment installed in home premises and offer a high economic viability.

Three-wheelers, too, come with the promise of a high economic viability on switching to electric. These vehicles can also make use of on-prem charging points in addition to distributed charging stations.

As we transition from three-wheeler light vehicles to four-wheelers and light and heavy goods vehicles, the viability of going electric comes down with increasing complexity of the vehicle. Four-wheelers, including passenger cars and taxis, have low route predictability but can make use of public charging points. Heavy goods vehicles have the least economic viability of them all. They must ferry goods across long distances and are often time-sensitive, which casts major doubts on their transition to EV.

Nevertheless, analysts are hopeful of a 25%-30% two-wheeler EV penetration and between 65% and 75% penetration of three-wheelers in India by 2030 while also expecting about 10%-12% of the bus fleet to be electrified by that time as long as governments — both at the Centre and the States — continue to incentivise EVs and standardise several pinpricks pertaining to component sourcing, manufacturing, and availability.

EV MANUFACTURERS IN INDIA

Back in 2001, the Maini Group introduced the first non-polluting compact electric vehicle, Reva, for the Indian market, which was also quite successful overseas. The company has been since taken over by Mahindra Electric Mobility Limited, but it is only now, after two decades, that we are seeing a burgeoning of EV companies on the market.

The e-Amrit portal indicates that there are 380 EV manufacturers in India. While detailing all of them is beyond the scope of this article, here are some popular OEMs and their current offerings:

Ather Energy

Ather Energy launched India's first smart scooter named S340 back in 2016. Early investments made by Ather in the EV segment allowed the company to expand its charging infrastructure to cover 21+ cities and 220 locations across India, which the company refers to as the Ather Grid. Ather currently offers the 450X and the 450 Plus EV scooters in India. These come with smart features such as Bluetooth connectivity, riding states, reverse assist, tyre pressure monitoring, cloud access, and more.

Currently, the Ather 450 Plus retails for ₹1,31,647 and the 450X for ₹1,50,657.

Tata Motors

Tata Motors is currently leading the EV four-wheeler segment in India with